

Exp No:28

COUNTING SEMAPHORE IN SYSTEMC

```
GNU nano 8.7.1 counting_semaphore.cpp *
#include <systemc.h>

SC_MODULE(counting_semaphore) {

    int sem;

    void task() {
        while(true) {
            wait(1, SC_SEC);

            if(sem > 0) {
                sem--;

                cout << "Task Accessing Resource at "
                     << sc_time_stamp()
                     << " Remaining: " << sem << endl;

                wait(2, SC_SEC);

                sem++;
            }
        }
    }

    SC_CTOR(counting_semaphore) {
        sem = 2;

        SC_THREAD(task);
        SC_THREAD(task);
    }
}

Write to File: counting_semaphore.cpp
^G Help      M-D DOS Format  M-A Append     M-B Backup File ^T Browse
^C Cancel    M-M Mac Format  M-P Prepend    M-Q Discard buffer
```

```
GNU nano 8.7.1 CMakeLists.txt *
cmake_minimum_required(VERSION 3.10)

project(CountingSemaphore)

set(CMAKE_CXX_STANDARD 17)

include_directories(/usr/include)

add_executable(counting_semaphore counting_semaphore.cpp)

target_link_libraries(counting_semaphore systemc)

Write to File: CMakeLists.txt
^G Help      M-D DOS Format  M-A Append     M-B Backup File ^T Browse
^C Cancel    M-M Mac Format  M-P Prepend    M-Q Discard buffer
```

OUTPUT:

```
sai_varshini@DESKTOP-5G7BJ: x + v
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Configuring done (4.7s)
-- Generating done (0.0s)
-- Build files have been written to: /home/sai_varshini/systemc_test/exp28/build
sai_varshini@DESKTOP-5G7BJ:~/systemc_test/exp28/build$ make
[ 50%] Building CXX object CMakeFiles/counting_semaphore.dir/counting_semaphore.cpp.o
[100%] Linking CXX executable counting_semaphore
[100%] Built target counting_semaphore
sai_varshini@DESKTOP-5G7BJ:~/systemc_test/exp28/build$ ./counting_semaphore

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
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Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_0
In file: kernel/sc_object_manager.cpp:153

Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_1
In file: kernel/sc_object_manager.cpp:153
Task Accessing Resource at 1 s Remaining: 1
Task Accessing Resource at 1 s Remaining: 0
Task Accessing Resource at 3 s Remaining: 0
Task Accessing Resource at 4 s Remaining: 0
Task Accessing Resource at 5 s Remaining: 0
Task Accessing Resource at 6 s Remaining: 0
Task Accessing Resource at 7 s Remaining: 0
Task Accessing Resource at 8 s Remaining: 0
Task Accessing Resource at 9 s Remaining: 0
sai_varshini@DESKTOP-5G7BJ:~/systemc_test/exp28/build$ |
```